

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title: COMPLETED RESEARCH		Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 15

(1) Research Program/Project/Study Title: Performance of Cucumber (<i>Cucumis sativus</i> L.) Applied with Different Soil Ameliorants	
Author's Name:	DR. SALVACION J. LEGASPI
Academic Rank:	PROFESSOR VI
Co-Authors (with academic rank):	HERALYN D. PADERES
Campus/Satellite College/Address:	CAPSU - BURIAS CAMPUS
Telephone/Cell Phone/Fax Number:	09171162490
E-mail Address:	csrdc@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		Mambusao	
Province:		Capiz	
Region:		VI	
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Cucumber		(8) Discipline/Field of Specialization Agriculture	

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 2 of 15

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted at the Organic Agriculture area of Capiz State University - Burias Campus, Mambusao, Capiz. It was laid out using a Randomized Complete Block Design (RCBD) for a two-factor experiment. Factor A consisted of different soil ameliorants, namely: A1 - Control, A2 - CAFSA with chicken dung, A3 - CAFSA with hog manure, A4 - Vermicast, and A5 - Ordinary compost. Factor B consisted of planting schedules such as B1 - 1st planting (October-November) and B2 - 2nd planting (December-January). All data gathered were subjected to Analysis of Variance (ANOVA) using STAR software. Differences among treatments were determined using Least Significance Difference (LSD) and Duncan Multiple Range Test (DMRT).

Growth and yield parameters of cucumber were not significantly affected by the application of different CAFSA except for the number of marketable fruits. Application of ordinary compost, CAFSA with hog manure, and vermicast gave the highest number of marketable fruits of cucumber. In terms of planting schedule, cucumbers during the first planting schedule exhibited a positive effect in all parameters except for the length of roots. A significant interaction among factors in influencing the number of marketable fruits was also noted.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

CAFSA, vermicast, planting schedule, hog manure, chicken dung

- 1. Introduction** (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Cucumber (*Cucumis sativus L.*) is a widely cultivated crop known for its high nutritional value and economic importance. The use of soil ameliorants can significantly enhance cucumber growth and yield by improving soil properties and nutrient availability. The Capiz State University (CAPSU) has developed a comprehensive soil ameliorant mix that includes animal manure, garden soil, compost, vermicast, rice bran, carbonized rice hull, effective microorganism inoculant, fermented plant juice, fish amino acid, indigenous microorganisms, lactic acid, molasses, and water.

Soil ameliorants are essential for enhancing soil fertility, structure, and biological activity, which are crucial for optimal plant growth and productivity (Zhang et al., 2013). The CAPSU formulated mix aims to address various soil deficiencies and provide a balanced nutrient supply to support cucumber growth.

Animal manure is a key organic amendment that improves soil fertility by supplying essential nutrients such as nitrogen, phosphorus, and potassium (Goss et al., 2013). It also enhances soil structure, water-holding capacity, and microbial activity (Chaudhry et al., 2009). Studies have shown that the application of animal manure can significantly increase cucumber yield and improve soil health (Nash et al., 2012).

Garden soil in the mix provides a stable base and contributes to the soil's mineral content and texture, supporting root development and nutrient uptake (Brady & Weil, 2002). Compost, a product of decomposed organic matter, enriches the soil with humus, enhancing nutrient availability and promoting beneficial microbial populations (Bernal et al., 2009). The use of compost in cucumber cultivation has been associated with improved growth parameters and yield (Meena et al., 2016).

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 3 of 15

Vermicast, or worm castings, is another valuable component known for its high nutrient content and ability to enhance soil structure and microbial activity (Arancon et al., 2005). Research indicates that vermicast application can improve cucumber plant health and increase yield by enhancing nutrient availability and soil biological properties (Lazcano & Domínguez, 2011).

Rice bran and carbonized rice hull are by-products of rice processing, rich in nutrients and organic matter. Rice bran provides essential minerals and organic compounds that support plant growth, while carbonized rice hull improves soil aeration and water retention (Sombroek et al., 1993). These materials have been shown to enhance soil fertility and crop performance (Yamamoto et al., 2014).

Effective microorganism (EM) inoculants, fermented plant juice, fish amino acid, indigenous microorganisms, lactic acid, and molasses are all components that contribute to the biological enhancement of the soil. EM inoculants introduce beneficial microbes that improve nutrient cycling and plant health (Higa & Parr, 1994). Fermented plant juice and fish amino acid provide organic nutrients and growth-promoting compounds, enhancing plant vigor and yield (Kopittke et al., 2019). Indigenous microorganisms and lactic acid promote a healthy soil microbial ecosystem, improving soil fertility and plant growth (Tamang et al., 2016). Molasses serves as a carbon source, supporting microbial activity and nutrient availability (Kahrl et al., 2010).

Water is the medium that facilitates the distribution and activation of these components in the soil. Proper irrigation practices ensure that the soil ameliorants are effectively integrated into the soil, providing consistent moisture and nutrient availability for optimal cucumber growth (Feres & Soriano, 2007).

The synergistic effect of these components in the CAPSU formulated soil ameliorants can lead to significant improvements in cucumber performance. Enhanced soil structure, increased nutrient availability, and improved microbial activity create an optimal environment for cucumber growth, resulting in higher yields and better-quality produce (Marschner, 2012).

The performance of cucumbers applied with CAPSU-formulated soil ameliorants is influenced by the comprehensive enhancement of soil properties. The components of formulated soil ameliorants work together to improve soil fertility, structure, and biological activity, leading to improved cucumber growth and yield.

Objectives of the study

The study was conducted to determine the performance of cucumber as affected by different soil ameliorants and planting schedules. Specifically, it aimed to:

1. Determine the growth and yield performance of cucumber applied with different soil ameliorants;
2. Evaluate which soil ameliorants were best for the growth and yield of cucumber;
3. Determine the effect of planting schedules on the growth and yield of cucumber; and
4. Determine the interaction effect between the application of different soil ameliorants and planting schedules.

(10) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The study was conducted at the Organic Agriculture area of Capiz State University -

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 4 of 15

Burias Campus, Mambusao, Capiz. It was laid out using a Randomized Complete Block Design (RCBD) for a two-factor experiment. Factor A consisted of different CAFSA Formulations namely: A1 - Control, A2 - CAFSA with chicken dung, A3 - CAFSA with hog manure, A4 - Vermicast, and A5 - Ordinary compost. Factor B consisted of planting schedules such as B1 - 1st planting (October-November) and B2 - 2nd planting (December-January). All data gathered were subjected to Analysis of Variance (ANOVA) using STAR software. Differences among treatments were determined using Least Significance Difference (LSD) and Duncan Multiple Range Test (DMRT).

(11) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Plant Height (cm)

The analysis of variance for the plant height of cucumber showed a highly significant effect of planting schedules and a not significant effect of the different CAFSA formulations. Cucumber grown during the 1st planting had the tallest height (263.02 cm), while the shortest plant height was obtained by cucumber grown during the second planting (255.57 cm). The mean of the plant height ranged from 132.74 cm to 309.11 cm for CAFSA. There was no significant interaction effect between the different CAFSA formulation and planting schedule.

Stem Girth (mm)

The ANOVA indicated that stem girth was highly affected by the planting schedule but not by the application of different CAFSA formulations. Cucumber produced during the first planting schedule had the biggest stem girth (6.33 mm) compared to the second planting schedule (4.67 mm). Additionally, stem girth of cucumber ranged from 5.29 mm to 5.66 mm when applied with different CAFSA formulations. The ANOVA further revealed that the two factors did not interact to influence the stem girth of cucumber.

Number of Productive Branches

The ANOVA for the number of productive branches showed a highly significant effect among the two planting schedules but a not significant result from the application of different CAFSA formulations. The mean ranged from 1.94 to 2.42 for different CAFSA formulations. Cucumber grown during the first planting had the highest number of productive branches with a mean of 3.03, while the lowest number of productive branches was obtained by cucumber grown during the second planting (1.37). There was no significant interaction effect between the application of different soil ameliorants and the season of cropping.

Number of Lateral Branches

The analysis of variance for the lateral branches revealed a highly significant effect on the planting schedule but not on the application of different soil ameliorants. In terms of the application of soil ameliorants, the means ranged from 4.70 to 5.25 branches. Fore planting schedule, crops planted during the second planting (21.27) produced more lateral branches then those grown during the first planting (7.47). There was no interaction effect between the application of different soil ameliorants and the season of cropping.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 15

Length of Roots

The ANOVA for the length of roots was not significantly affected either by the application of different CAFSA formulations or the planting schedules. In terms of the application of different CAFSA formulations, the mean ranged from 27.56 cm to 29.39 cm. Among the planting schedules, the mean ranged from 25.41 cm to 30.11 cm. The ANOVA further revealed that the two factors did not interact to influence the root length of cucumber.

Biomass

The analysis of variance for the biomass revealed a highly significant effect on the planting schedule but a not significant effect on the application of CAFSA formulations. For the CAFSA, the mean ranged from 162.34 g to 213.17 g. For the planting schedule, cucumber grown during the first planting schedule had the heaviest weight with a mean of 224.99 g, whereas cucumber grown during the second planting schedule obtained the lightest weight with a mean of 169.67 g. No significant interaction effect between the two factors was reflected in the result of ANOVA.

Length of Fruit (cm)

The ANOVA for the length of fruit showed a highly significant effect of planting schedules but not with the application of different CAFSA formulations. For the application of CAFSA formulations, the mean ranged from 19.79 cm to 20.40 cm. For the planting schedule, cucumber produced during the first planting obtained the longest fruit length (21.29 cm), while the shortest fruit was obtained by cucumber planted during the second planting (18.91 cm). There was no interaction effect between the CAFSA formulation and planting schedules in influencing the length of fruit of cucumber.

Diameter of Fruit (mm)

The analysis of variance for the diameter of fruit revealed a highly significant effect of the planting schedule but not the application of different CAFSA formulations. For the CAFSA formulations, the mean ranged from 44.62 mm to 46.64 mm. For the effect of planting schedule, cucumber planted during the first planting has the biggest diameter of fruit (47.04 mm). The smallest fruit was obtained by cucumber planted during the second planting (43.90 mm). There was no interaction effect between the CAFSA formulations and planting schedules.

Number of Marketable Fruits

The different soil ameliorants and planting schedules significantly affected the number of marketable fruits of cucumber. A highly significant interaction between the factors in influencing the number of marketable fruits was observed. The most marketable fruits were noted in the cucumber applied with ordinary compost in the 1st planting schedule (6.67), which was comparable to cucumber when applied with CAFSA with hog manure (6.50) and ordinary compost (6.50) in the 2nd planting, and vermicast in the 1st planting schedule (6.38). This was followed by CAFSA with chicken dung in both planting schedules (6.17 and 6.13), cucumber with no application (6.13), CAFSA with hog manure (6.20) in the 1st planting, and vermicast in the 2nd planting schedule (6.07). The least number of fruits was obtained by cucumber with no application grown during the 2nd planting schedule (5.4).

Weight of Marketable Fruits (g)

The analysis of variance for the weight of marketable fruits revealed a highly

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 6 of 15

significant effect of the planting methods but not with the application of different CAFSA. The heaviest weight of marketable fruits was observed from cucumber grown during the first planting schedule with a mean of 269.76 g, while the lightest weight of harvested marketable fruits was from cucumber grown during the second planting schedule with a mean of 198.32 g. For the effect of CAFSA formulations, the mean ranged from 198.32 g to 269.76 g. No interaction effect between the applications of different soil ameliorants and planting schedules was observed.

Yield in tons per hectare

The ANOVA for the yield in tons per hectare showed a highly significant effect of the planting schedule but a not effect of the application of different CAFSA formulations. For the application of different soil ameliorants, the mean ranged from 0.59 tons/ha to 0.88 tons/ha. Between the planting schedules, the highest yield was obtained by cucumber planted during the first planting (0.82 tons/ha), while the lowest yield was observed from cucumber grown during the second planting (0.60 tons/ha). There was no interaction effect between the CAFSA formulations and planting schedule.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the results of the study, the following conclusions were drawn:

1. The growth and yield of cucumber were not influenced by the application CAFSA with different formulations except the number of marketable fruits.
2. Planting schedule affected all the growth parameters except the length of roots.
3. Growth and yield parameters of cucumber were highly significantly affected by the planting schedule, except for the length of the roots.
4. There was an interaction effect between the CAFSA formulations and planting schedules in influencing the number of marketable fruits of cucumber.
5. Cucumber produced better yield when planted in October to November.

Recommendations

Based on the results of the study, the following recommendations are hereby forwarded:

1. Grow cucumber during the month of October to November for better growth and yield.
2. Use any of the organic soil ameliorants and CAFSA formulations in growing cucumber.
3. Use any of the soil ameliorants in growing cucumber.
4. Conduct similar studies to further assess the effectiveness of different CAFSA formulations on other crops.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 7 of 15

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

- Arancon, N. Q., Edwards, C. A., Atiyeh, R., & Metzger, J. D. (2005). Effects of vermicomposts produced from food waste on the growth and yields of greenhouse peppers. *Bioresource Technology*, 93(2), 139-144.
- Bernal, M. P., Alburquerque, J. A., & Moral, R. (2009). Composting of animal manures and chemical criteria for compost maturity assessment. A review. *Bioresource Technology*, 100(22), 5444-5453.
- Brady, N. C., & Weil, R. R. (2002). *The Nature and Properties of Soils* (13th ed.). Prentice Hall.
- Chaudhry, A. N., Khan, K. S., & Khaliq, A. (2009). The effect of animal manure and chemical fertilizer on the soil physical properties and NPK contents. *Soil & Environment*, 28(2), 113-121.
- Fereres, E., & Soriano, M. A. (2007). Deficit irrigation for reducing agricultural water use. *Journal of Experimental Botany*, 58(2), 147-159.
- Goss, M. J., Tubeileh, A., & Goorahoo, D. (2013). A review of the use of organic amendments and the risk to human health. *Advances in Agronomy*, 120, 275-379.
- Higa, T., & Parr, J. F. (1994). Beneficial and effective microorganisms for a sustainable agriculture and environment. International Nature Farming Research Center, Atami, Japan.
- Kahrl, F., Li, Y., Su, Y., Tennigkeit, T., Wilkes, A., & Xu, J. (2010). Greenhouse gas emissions from nitrogen fertilizer use in China. *Environmental Science & Policy*, 13(8), 688-694.
- Kopittke, P. M., Menzies, N. W., Wang, P., McKenna, B. A., & Lombi, E. (2019). Soil and the intensification of agriculture for global food security. *Environment International*, 132, 105078.
- Lazcano, C., & Domínguez, J. (2011). The use of vermicompost in sustainable agriculture: Impact on plant growth and soil fertility. *Soil Ecology Research Developments*, 10, 1-23.
- Marschner, P. (2012). *Marschner's Mineral Nutrition of Higher Plants*. Elsevier.
- Meena, R. S., Meena, V. S., Verma, J. P., & Kumar, A. (2016). The needs of healthy soils for a healthy world. *Journal of Cleaner Production*, 112, 4594-4595.
- Nash, P. R., Bradley, K. W., & Murrell, T. S. (2012). Animal manure as a nutrient source for peanut production. *Peanut Science*, 39(1), 9-15.
- Sombroek, W. G., Nachtergaele, F. O., & Hebel, A. (1993). Amounts, dynamics and sequestering of carbon in tropical and subtropical soils. *Ambio*, 22(7), 417-426.
- Tamang, J. P., Watanabe, K., & Holzapfel, W. H. (2016). Review: Diversity of microorganisms in global fermented foods and beverages. *Frontiers in Microbiology*, 7, 377.
- Yamamoto, A., Ishii, Y., & Koizumi, M. (2014). Effects of carbonized rice hull application on rice seedling establishment and growth performance in nursery boxes. *Plant Production Science*, 17(3), 264-273.
- Zhang, X., Davidson, E. A., Mauzerall, D. L., Searchinger, T. D., Dumas, P., & Shen, Y. (2013). Managing nitrogen for sustainable development. *Nature*, 528(7580), 51-59.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 8 of 15

APPENDICES

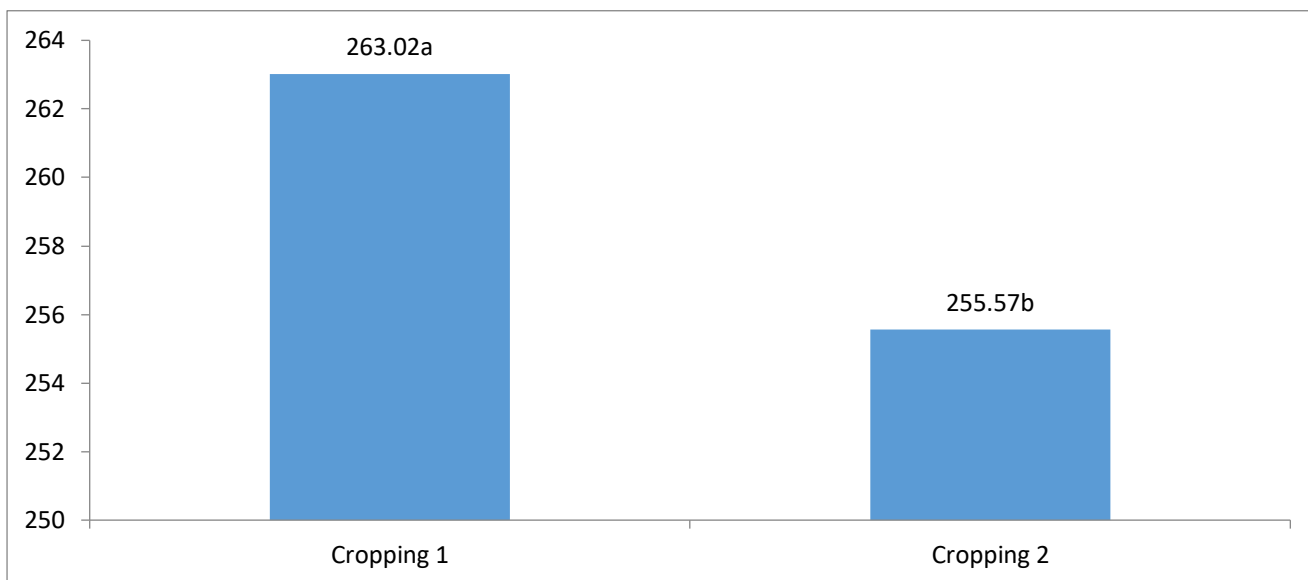


Figure 1. Plant height (cm) of cucumber applied with different soil ameliorant as affected by two cropping seasons.

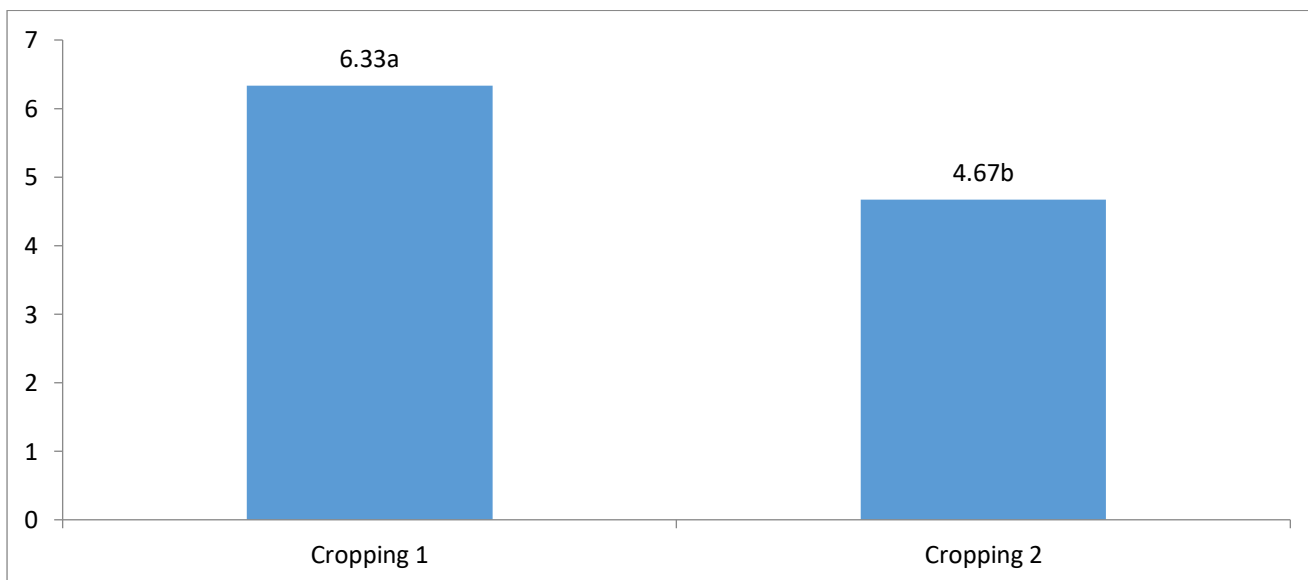


Figure 2. Stem girth (mm) of cucumber applied with different soil ameliorant affected by two cropping seasons.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 9 of 15

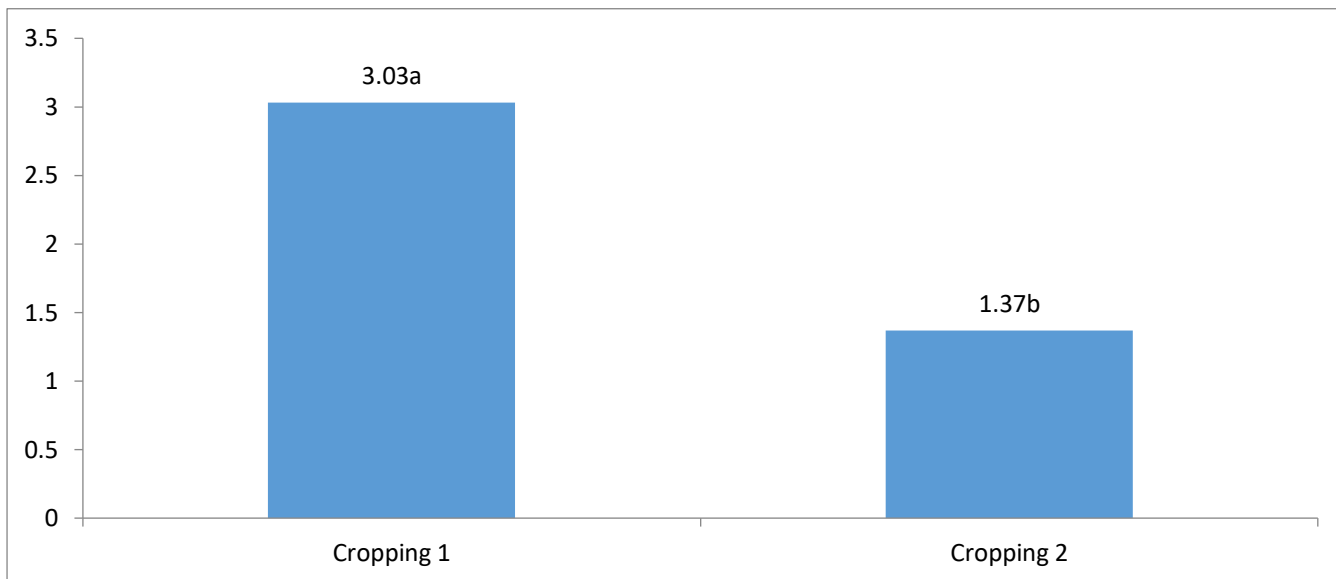


Figure 3. Number of productive branches of cucumber applied with different soil ameliorant as affected by two cropping seasons.

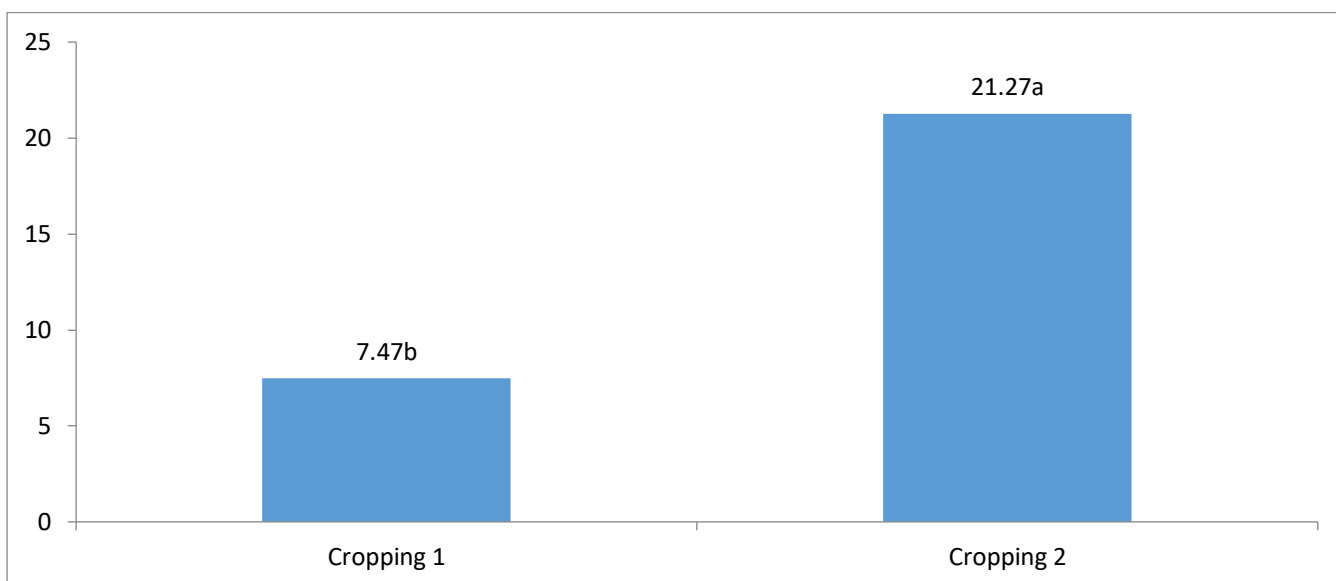


Figure 4. Lateral branches of cucumber applied with different soil ameliorant as affected by two cropping seasons.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 10 of 15

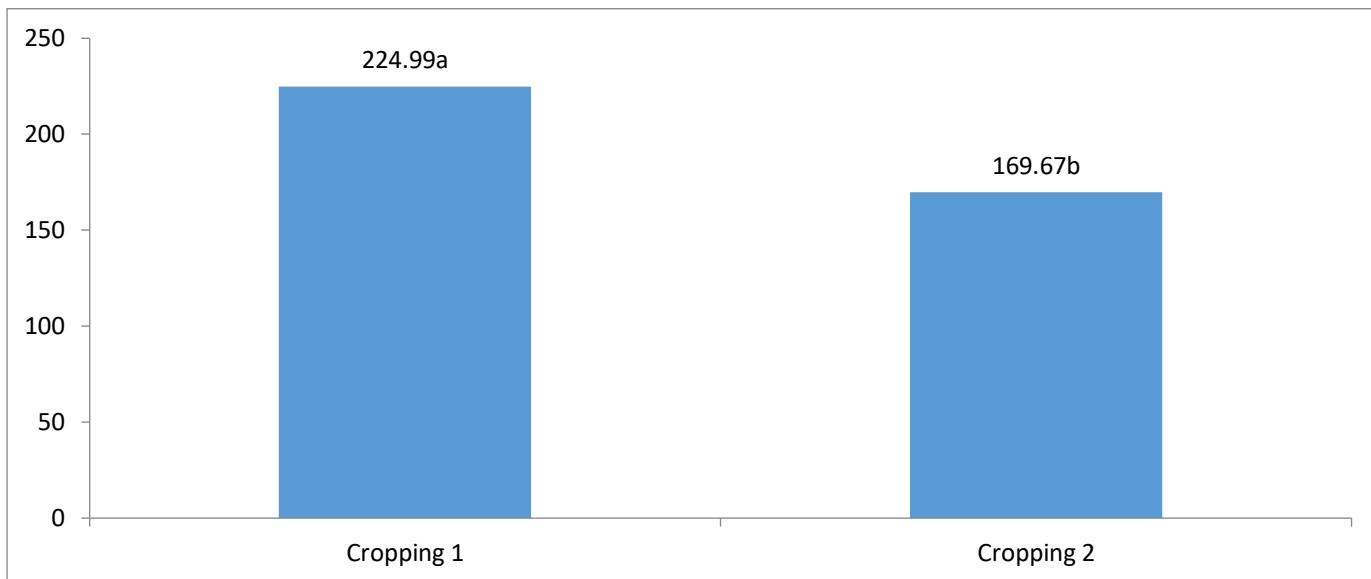


Figure 5. Biomass (g) of cucumber applied with different soil ameliorant as affected by two cropping seasons.

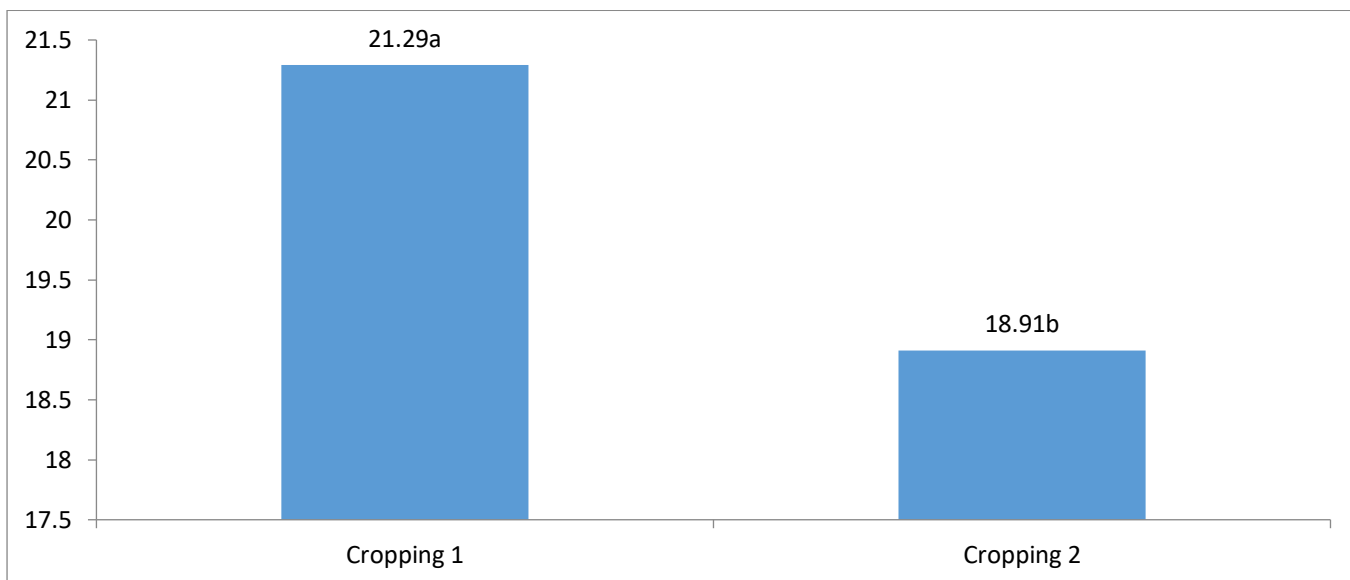


Figure6. Length of fruit of cucumber applied with different soil ameliorant affected by two cropping seasons.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 11 of 15

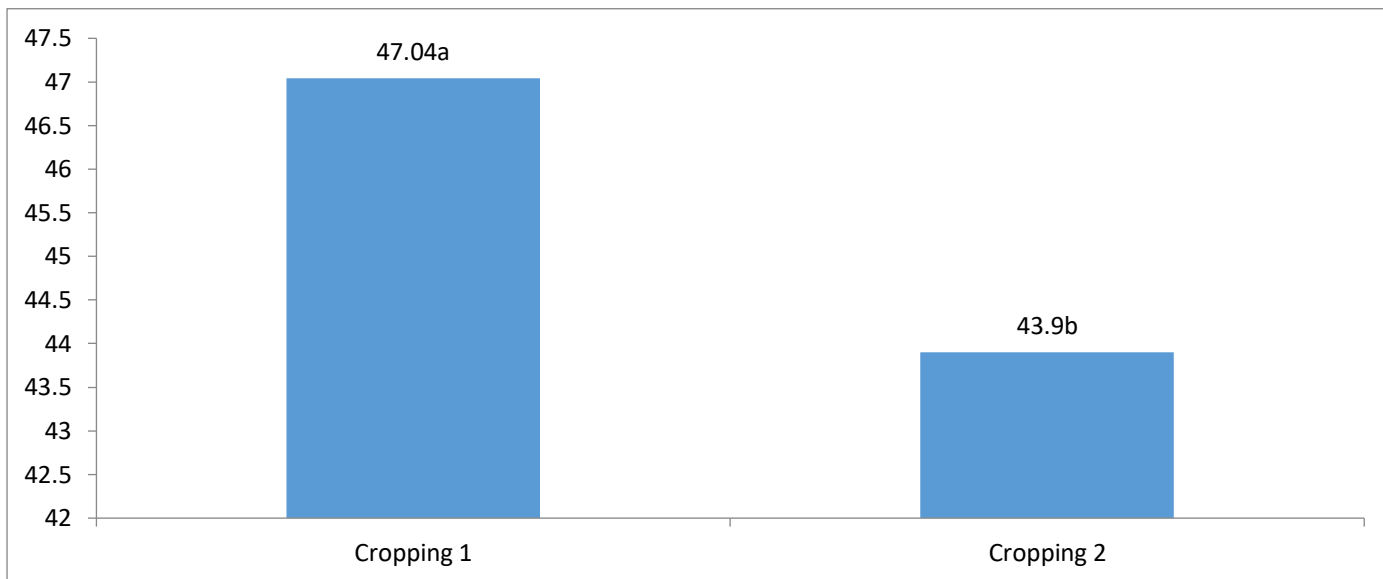


Figure 7. Diameter of fruit of cucumber applied with different soil ameliorant affected by two cropping seasons.

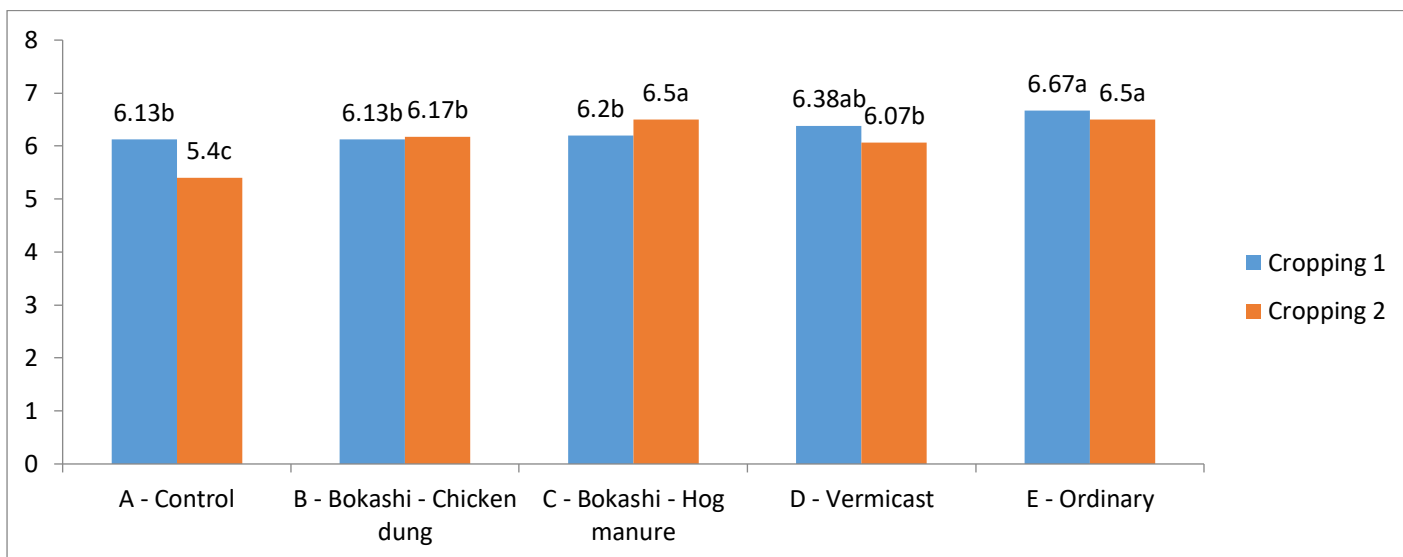


Figure 8. Interaction effect between the different soil ameliorants and cropping seasons on the number of marketable fruit in cucumber.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 12 of 15

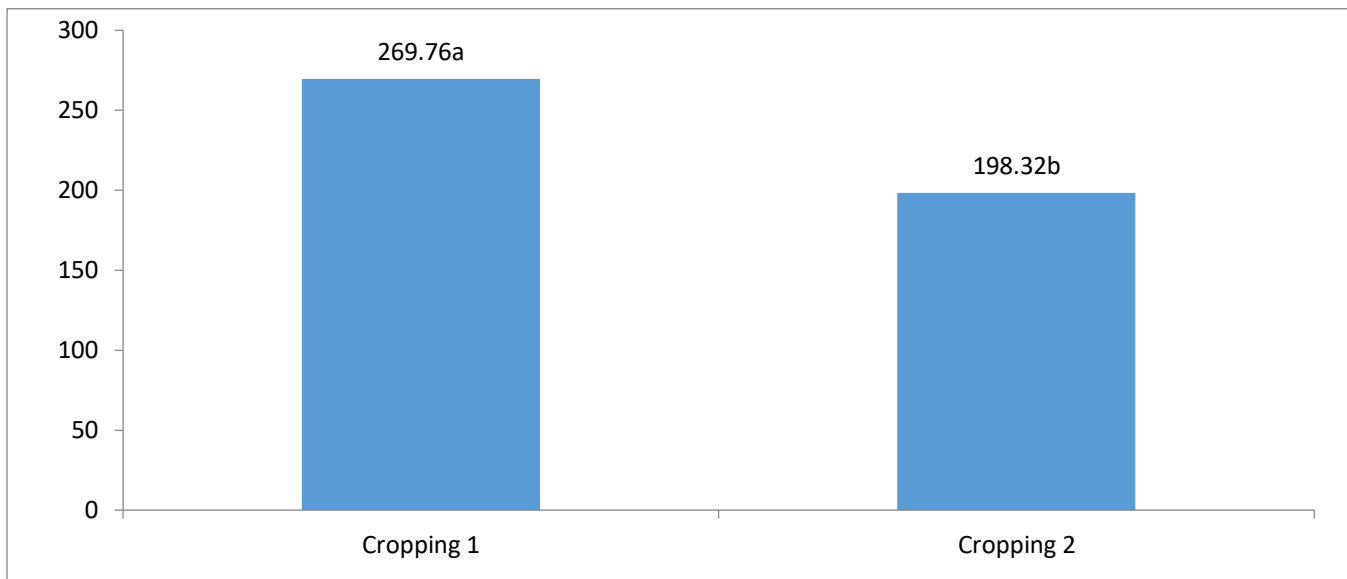


Figure9. Weight (g) of marketable fruit in cucumber applied with different soil ameliorant affected by two cropping seasons.

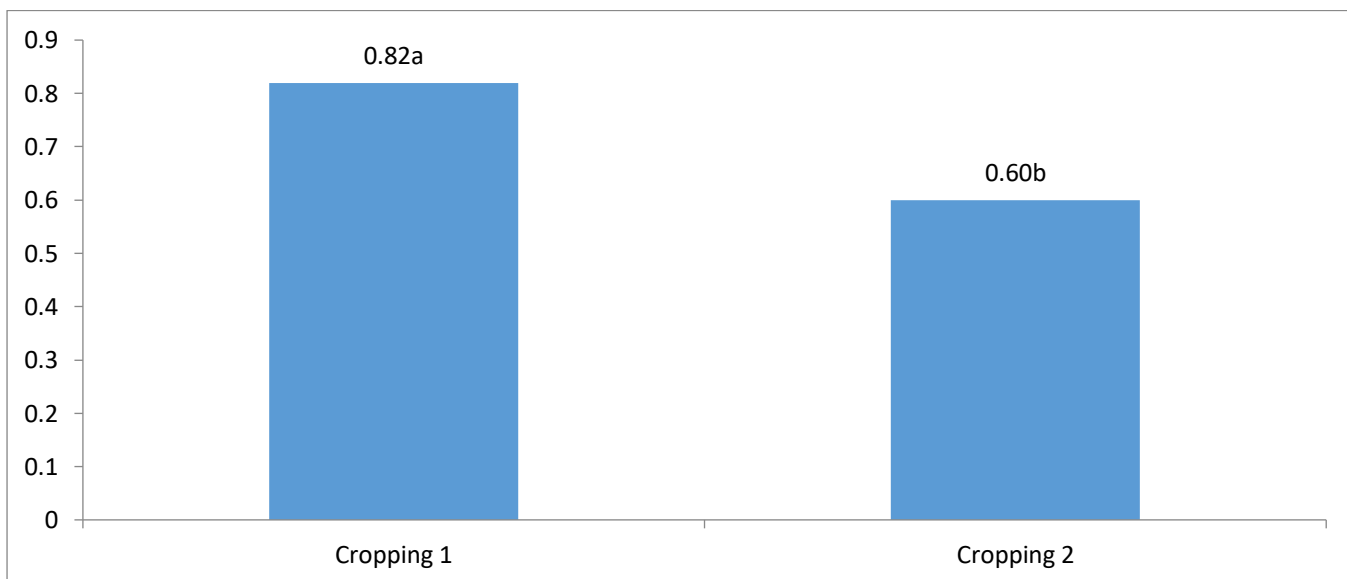


Figure10. Yield of cucumber applied with different soil ameliorant affected by two cropping seasons.

	Document Type:	Document Code	RES-F08
	FORM ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	January 26, 2023
	COMPLETED RESEARCH	Page	Page 1 of 15

Table 1. Plant height (cm) and stem girth (mm) of cucumber (*Cucumis sativus* L.) applied with different soil ameliorant in two cropping.

Plant Height (cm)					Stem girth (mm)				
Treatment	Cropping 1	Cropping 2	Total	Mean	Treatment	Cropping 1	Cropping 2	Total	Mean
A - Control	27.15	238.32	265.47	132.74	A - Control	5.87	4.7	10.57	5.29
B - Bokashi with Chicken dung	348.11	270.11	618.22	309.11	B - Bokashi - Chicken dung	6.67	4.37	11.04	5.52
C - Bokashi with Hog manure	290.61	270.32	560.93	280.47	C - Bokashi - Hog manure	6.07	4.7	10.77	5.39
D - Vermicast	311.52	251.97	563.49	281.75	D - Vermicast	6.32	4.97	11.29	5.65
E - Ordinary	337.7	247.15	584.85	292.43	E - Ordinary	6.71	4.6	11.31	5.66
Total	1315.09	1277.87			Total	31.64	23.34		
Mean	263.02a	255.57b			Mean	6.33a	4.67b		

Table 2. Number of productive and lateral branches of cucumber (*Cucumis sativus* L.) applied with different soil ameliorant in two cropping.

Number of Productive Branches					Number of Lateral Branches				
Treatment	Cropping 1	Cropping 2	Total	Mean	Treatment	Cropping 1	Cropping 2	Total	Mean
A - Control	2.4	1.47	3.87	1.94	A - Control	7	19.33	26.33	13.17
B - Bokashi - Chicken dung	3.6	1.23	4.83	2.42	B - Bokashi - Chicken dung	8.67	22.67	31.34	15.67
C - Bokashi - Hog manure	2.97	1.57	4.54	2.27	C - Bokashi - Hog manure	7.67	24.67	32.34	16.17
D - Vermicast	2.87	1.07	3.94	1.97	D - Vermicast	6.67	18.67	25.34	12.67
E - Ordinary	3.3	1.5	4.8	2.40	E - Ordinary	7.33	21	28.33	14.17
Total	15.14	6.84			Total	37.34	106.34		
Mean	3.03a	1.37b			Mean	7.47b	21.27a		

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 2 of 15

Table 3. Length of roots (cm) and biomass (g) of cucumber (*Cucumis sativus* L.) applied with different soil ameliorant in two cropping.

Length of Roots ^{ns}					Biomass				
Treatment	Cropping 1	Cropping 2	Total	Mean	Treatment	Cropping 1	Cropping 2	Total	Mean
A - Control	27.06	28.05	55.11	27.56	A - Control	175.2	149.47	324.67	162.34
B - Bokashi - Chicken dung	35.84	22.94	58.78	29.39	B - Bokashi - Chicken dung	272.87	169.33	442.2	221.10
C - Bokashi - Hog manure	26.93	28.91	55.84	27.92	C - Bokashi - Hog manure	203.3	182.87	386.17	193.09
D - Vermicast	31.9	25.14	57.04	28.52	D - Vermicast	222.87	169.7	392.57	196.29
E - Ordinary	28.84	22	50.84	25.42	E - Ordinary	250.73	177	427.73	213.87
Total	150.57	127.04			Total	1124.97	848.37		
Mean	30.11	25.41			Mean	224.99a	169.67b		

Table 4. Length of fruit and diameter of cucumber (*Cucumis sativus* L.) applied with different soil ameliorant in two cropping.

Length of Fruit					Diameter of Fruit				
Treatment	Cropping 1	Cropping 2	Total	Mean	Treatment	Cropping 1	Cropping 2	Total	Mean
A - Control	21.22	19.3	40.52	20.26	A - Control	45.77	43.47	89.24	44.62
B - Bokashi - Chicken dung	21.87	18.92	40.79	20.40	B - Bokashi - Chicken dung	49.51	43.77	93.28	46.64
C - Bokashi - Hog manure	20.76	18.81	39.57	19.79	C - Bokashi - Hog manure	46.07	43.96	90.03	45.02
D - Vermicast	20.55	19.95	40.5	20.25	D - Vermicast	46.48	44.31	90.79	45.40
E - Ordinary	22.07	17.56	39.63	19.82	E - Ordinary	47.37	43.98	91.35	45.68
Total	106.47	94.54			Total	235.20	219.49		
Mean	21.29a	18.91b			Mean	47.04a	43.90b		

Table 5. Number and weight (g) of marketable fruit in cucumber (*Cucumis sativus* L.) applied with different soil ameliorant in two cropping.

	Document Type:	Document Code	RES-F08
	FORM ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	January 26, 2023
	COMPLETED RESEARCH	Page	Page 3 of 15

Number of marketable fruit **					Wt. of Marketable Fruit				
Treatment**	Cropping 1	Cropping 2	Total	Mean	Treatment	Cropping 1	Cropping 2	Total	Mean
A - Control	6.13b	5.4c	11.53	5.77	A - Control	249.51	192.31	441.82	220.91
B - Bokashi - Chicken dung	6.13b	6.17b	12.3	6.15	B - Bokashi - Chicken dung	282.29	195.41	477.7	238.85
C - Bokashi - Hog manure	6.2b	6.5a	12.7	6.35	C - Bokashi - Hog manure	261.49	209.13	470.62	235.31
D - Vermicast	6.38ab	6.07b	12.45	6.23	D - Vermicast	268.17	203	471.17	235.59
E - Ordinary	6.67a	6.5a	13.17	6.59	E - Ordinary	287.32	191.77	479.09	239.55
Total	31.51	30.64			Total	1348.78	991.62		
Mean	6.30	6.13			Mean	269.76a	198.32b		

Table 6. Yield of cucumber (*Cucumis sativus* L.) applied with different soil ameliorant in two cropping.

Yield (T/ha)				
Treatment	Cropping 1	Cropping 2	Total	Mean
A - Control	0.67	0.52	1.18	0.59
B - Bokashi - Chicken dung	0.80	0.58	1.38	0.69
C - Bokashi - Hog manure	0.70	0.6	1.38	0.69
D - Vermicast	1.12	0.64	1.76	0.88
E - Ordinary	1.02	0.64	1.66	0.83
Total	3.29	3.05		
Mean	0.82a	0.60b		